

Ex. No 5 Use Autopsy to create a case and import evidence

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link: https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

Download files: [4Dell Latitude CPi.E01](#), [4Dell Latitude CPi.E02](#)

1. Installation

- **Download and Install:** Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

- **Open Autopsy:** Launch the application after installation.
- **Create a New Case:**
 - Click on New Case.
 - Enter the case name and location where the case data will be stored.
 - Fill in the details like the case number, examiner's name, etc., and click Next.

3. Adding a Data Source

- **Choose the Type of Data Source:**
 - After creating a case, you'll be prompted to add a data source.
 - You can add disk images, directories, logical files, or local disks.
- **Select the Data Source:**
 - Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
 - https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- **Configure Ingest Modules:**
 - Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
 - You can enable or disable these based on your investigation needs.
- **Start Analysis:** Click Next to start the analysis.

4. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.
- **Explore the Resulting Artifacts:**
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.

- **Use the Tree Viewer:**
 - On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- **Keyword Search:**
 - You can perform specific keyword searches using the Keyword Search module.
 - Use pre-configured lists or enter custom keywords.
- **File Analysis:**
 - Navigate through files and folders under the File Types or File System section.
 - Open, view, or export files for further examination.
- **Timeline Analysis:**
 - Use the Timeline module to visualize events based on timestamps.
 - This can help track user activity over time.
- **Hash Analysis:**
 - Compare file hashes with known databases to identify known good or bad files.

6. Reporting

- **Generate a Report:**
 - After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
 - Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.

7. Case Closure

- **Close the Case:**
 - Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**
 - Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional)

- **Custom Ingest Modules:**
 - Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment.

Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: 001

Examiner

Name: vamshi

Phone: 8341248633

Email:

Notes: Dell Latitude

Organization

Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

Add Data Source

Steps

1. **Select Host**
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Host

Hosts are used to organize data sources and other data.

☒ Generate new host name based on data source name

☐ Specify new host name

☐ Use existing host

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Add Data Source

Steps

1. Select Host
2. **Select Data Source Type**
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Data Source Type

☒

 Disk Image or VM File

☐

 Local Disk

☐

 Logical Files

☐

 Unallocated Space Image File

☐

 Autopsy Logical Imager Results

☐

 XRY Text Export

< Back Next > Finish Cancel Help

Add Data Source

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Steps

1. Select Host

2. Select Data Source Type

3. **Select Data Source**

4. Configure Ingest

5. Add Data Source

Select Data Source

Path:

D:\4Dell Latitude CPI(1).E01

Browse

☐

Ignore orphan files in FAT file systems

Time zone:

(GMT+5:30) Asia/Calcutta

Sector size:

Auto Detect

Bitlocker Password (optional):

Hash Values (optional):

MD5:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back

Next >

Finish

Cancel

Help

Add Data Source

×

Steps

1. Select Host

2. Select Data Source Type

3. Select Data Source

4. **Configure Ingest**

5. Add Data Source

Configure Ingest

Run ingest modules on:

All Files, Directories, and Unallocated Space

☒ Recent Activity

☒ Hash Lookup

☒ File Type Identification

☒ Extension Mismatch Detector

☒ Embedded File Extractor

☒ Picture Analyzer

☐ Keyword Search

☒ Email Parser

☒ Encryption Detection

☒ Interesting Files Identifier

☒ Central Repository

☒ PhotoRec Carver

☒ Virtual Machine Extractor

Select All

Deselect All

History

The selected module has no per-run settings.

Extracts recent user activity, such as Web browsing, rec..

Global Settings

< Back

Next >

Finish

Cancel

Help

